AI & ML Internship: Week 1 Learning Notes

Codomax Digital Solutions | AI & ML Intern

1. Basics of Artificial Intelligence and Machine Learning

Artificial Intelligence (AI) is the ability of a machine to perform tasks that normally need human intelligence, such as understanding language, recognizing images, solving problems and making decisions.

Machine Learning (ML) is a part of AI where a computer learns patterns from data instead of being given fixed rules. The more good data it sees, the better it gets.

Types of Machine Learning

1. **Supervised learning:** learns from labeled data (input with the correct answer). Examples: spam detection, house price prediction, disease classification.
2. **Unsupervised learning:** finds patterns in unlabeled data. Examples: customer segmentation, grouping similar news articles.
3. **Reinforcement learning:** an agent learns by trial and error and gets rewards or penalties. Examples: game-playing AI, robotics, self-driving systems.

Basic ML Workflow

1. Collect data
2. Clean and prepare data
3. Split into training and testing sets
4. Train a model
5. Evaluate it on test data
6. Improve and deploy

Key Terms

- **Dataset:** the collection of data used to train and test.
- **Feature:** an input variable (e.g., house size).
- **Label:** the output we want to predict (e.g., house price).
- **Model:** the learned pattern or function that makes predictions.
- **Training:** the process of the model learning from data.
- **Overfitting:** the model memorizes training data and performs badly on new data.

2. AI vs. Machine Learning vs. Deep Learning

	AI	Machine Learning	Deep Learning
Meaning	Making machines act intelligently	Machines learn from data	ML using multi-layer neural networks
Scope	Broadest	Subset of AI	Subset of ML
Data needed	Varies	Moderate	Very large
Example	Chatbot, rule-based expert system	Recommendation engine, spam filter	Face recognition, speech recognition, language models

Remember: Deep Learning ⊂ Machine Learning ⊂ Artificial Intelligence (nested circles).

Neural network in one line: layers of connected "neurons" that pass numbers forward, adjusting their weights during training to reduce errors.

3. Real-World AI Applications

- **Healthcare:** detecting diseases from X-rays and scans, drug discovery.
- **Finance:** fraud detection, credit scoring, stock trend analysis.
- **E-commerce:** product recommendations on Amazon and Flipkart.
- **Entertainment:** Netflix and Spotify suggestions, content creation.
- **Transport:** self-driving cars, traffic prediction, route optimization on Google Maps.
- **Agriculture:** crop disease detection, yield prediction, smart irrigation.
- **Education:** personalized learning, automated grading.
- **Daily life:** voice assistants (Siri, Alexa, Google Assistant), face unlock, Google Translate, email spam filters.

How to explain one in an answer: state the problem, say what data the AI uses, and describe the benefit. Example: *Netflix has millions of titles and users cannot browse all of them. It uses your watch history to predict what you will enjoy, which saves time and keeps users engaged.*

4. Installing Python

1. Go to python.org and download the latest stable version.
2. Run the installer. On Windows, tick **Add Python to PATH**.

3. Open a terminal or command prompt and run `python --version` to verify.
4. Use an editor such as VS Code or IDLE, or use Google Colab to code in the browser with no setup.
5. Run a file with `python filename.py`.

5. Python Programming Basics

Variables and Data Types

A variable stores a value. Python figures out the type automatically.

```
name = "Raviteja"     # str
age = 21              # int
height = 5.9          # float
is_student = True     # bool
print(name, age)
print(type(age))
```

Operators

- Arithmetic: `+ - * / // % **`
- Comparison: `== != > < >= <=`
- Logical: `and or not`

Input and Output

```
name = input("Enter your name: ")
print("Hello,", name)
```

`input()` always returns a string, so use `int()` or `float()` to convert numbers.

Conditions

```
marks = 72
if marks >= 90:
    print("A")
elif marks >= 60:
    print("B")
else:
    print("C")
```

Loops

```
# for loop
for i in range(1, 6):
    print(i)

# while loop
count = 1
while count <= 3:
    print(count)
    count += 1
```

Use `break` to exit a loop early and `continue` to skip to the next round.

Lists

A list is an ordered, changeable collection.

```
skills = ["Python", "ML", "SQL"]
skills.append("Deep Learning")   # add
skills.remove("SQL")             # remove
print(skills[0])                 # first item
print(len(skills))               # length
for s in skills:
    print(s)
```

Functions

A function is a reusable block of code.

```
def add(a, b):
    return a + b

print(add(3, 4))   # 7
```

6. Beginner Programs to Practice

```
# 1. Even or odd
num = int(input("Enter a number: "))
print("Even" if num % 2 == 0 else "Odd")

# 2. Multiplication table
n = int(input("Table of: "))
for i in range(1, 11):
```

```
    print(f"{n} x {i} = {n*i}")

# 3. Largest number in a list
nums = [4, 19, 7, 33, 2]
print("Largest:", max(nums))

# 4. Reverse a string
print("Machine Learning"[::-1])

# 5. Factorial
def factorial(n):
    result = 1
    for i in range(1, n + 1):
        result *= i
    return result
print(factorial(5))

# 6. Simple calculator
def calculate(a, b, op):
    if op == "+": return a + b
    if op == "-": return a - b
    if op == "*": return a * b
    if op == "/": return a / b if b != 0 else "Cannot divide by zero"
print(calculate(10, 5, "*"))
```

7. Quick Revision Points

- AI is the goal, ML is the method, Deep Learning is a powerful technique within ML.
- ML learns from data; supervised uses labels, unsupervised does not, reinforcement uses rewards.
- Good data matters more than complex models.
- Python is popular for AI because it is simple and has libraries like NumPy, pandas, scikit-learn, TensorFlow and PyTorch.
- Practice by typing the code yourself, not just reading it.

8. Self-Check Questions

1. What is the difference between AI, ML and Deep Learning?
2. Name the three types of machine learning with one example each.
3. What is overfitting?
4. Give three real-world AI applications and explain one in detail.

5. What does `range(1, 6)` produce?
6. What is the difference between a list and a variable?
7. Why do we use functions?